



Microbial community / environmental DNA analysis with Chipster

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Schedule

Day 1: Tuesday 8.4. (9-13) **Amplicon data processing**

Day 2: Wednesday 9.4. (9-13) **Microbial community data analysis**

Tips for the course



- You can ask questions anytime
 - Use your microphone
 - Write your question in the course Google doc: bit.ly/eDNA2025
 - If you get stuck in the exercises, ask for help
- When you are not speaking, please mute your microphone
- Introductory round to test your mics: Please tell
 - Your name
 - Institute
 - If you have microbial community / eDNA data already and how have you analyzed it
 - What is the best thing about spring

What will I learn?

- Microbial community / environmental DNA analysis of amplicon sequencing data
 - Central concepts
 - Analysis steps
 - File formats
- Microbial 16S rRNA gene data is used in the exercises
 - We discuss also how to analyze data on other marker genes
- How to operate the Chipster software

Understanding data analysis - why?

- Bioinformaticians might not always be available when needed
- Biologists know their own experiments best
 - Potential batch effects etc
- Allows you to design experiments better → less money wasted
- Allows you to discuss more easily with bioinformaticians

Introduction to Chipster: You will learn



- What Chipster is
- How the user interface works
- How to import data
- How to store and share your analysis
- How to speed up your analysis
 - run the same analysis step for all the samples simultaneously
- How to get help
- Where to find training materials

Introduction to Chipster

- User-friendly analysis software for high-throughput data
- Provides an easy access to over 500 analysis tools
 - Command line tools
 - R/Bioconductor packages
- Free, open source software
- What can I do with Chipster?
 - analyze high-throughput data
 - visualize data efficiently
 - share analysis sessions



Chipster

Open source platform for data analysis



Welcome to Chipster

Chipster is a user-friendly analysis software for high-throughput data such as Visium, single-cell and bulk RNA-seq. Chipster provides a web interface to over 500 analysis tools, and the actual analysis jobs run on the server side making use of CSC's computing environment (watch [introduction video](#)).

If you would like to use Chipster hosted by CSC, you need a [user account](#). Please note that Chipster is also available for [local server installation](#) free of charge.

- Home
- Getting access
- Manual
- Tutorial videos
- Course material
- Cite
- Contact



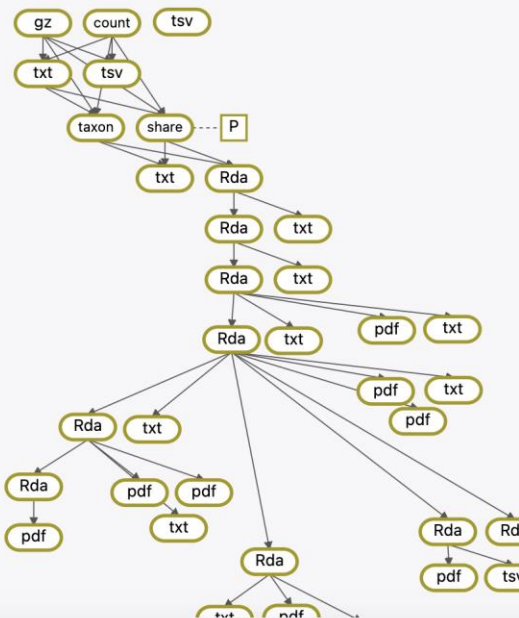
Launch Chipster

Training:

- 4.-5.3.2025 [Single-cell RNA-seq data analysis](#)

Tools

Find file



Toolset

Tool

- Quality control
- Preprocessing
- Utilities
- Matching sets of genomic regions
- Alignment
- Variants
- RNA-seq
- Small RNA-seq
- Single-cell RNA-seq (Seurat v5, new)
- Single-cell RNA-seq (Seurat v4, old)
- Spatially resolved transcriptomics (Seurat v5, new)
- Spatially resolved transcriptomics (Seurat v4, old)
- ChIP- and DNase-seq
- Microbial amplicon data preprocessing for ASV
- Microbial amplicon data preprocessing for OTU
- Microbial amplicon data analyses
- CNA-seq

 Find tool

Read quality with MultiQC for many FASTQ files

Read quality with MultiQC for input data bigger than 200 GB

Read quality with FastQC

Read quality statistics with FASTX

RNA-seq quality metrics with RSeQC

RNA-seq strandedness inference with RSeQC

Collect multiple metrics from BAM

PCA and heatmap of samples with DESeq2

Check FASTQ file for errors

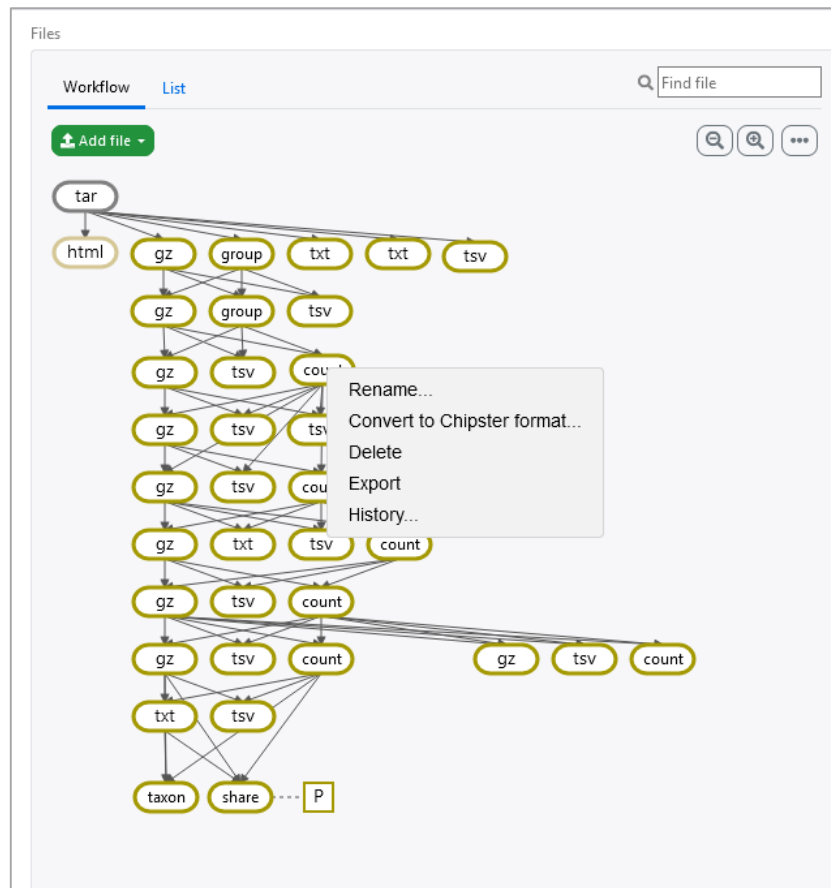
Parameters

Run

The tool runs FastQC on multiple FASTQ files, and then combines the reports using MultiQC. Input can be FASTQ files or tar files containing FASTQ files. Files can be gzipped. Please make sure you don't have duplicate FASTQ file names. Run the tool once for all samples, not separately for each file. This tool is based on the FastQC and MultiQC packages. [More info...](#)

Workflow view

- Shows the relationships of the files
- You can move the boxes around
- Several files can be selected by
 - keeping the Ctrl key down
 - drawing a box around them
- Right click allows you to
 - download a file ("Export")
 - delete a file
 - view analysis history



Analysis sessions

- Your analysis is saved automatically in the cloud
 - Session includes all the files, their relationships and metadata (what tool and parameters were used to produce each file).
 - Session is a single .zip file.
 - Note that cloud sessions are not stored forever! Remember to download the session when ready.
- You can share sessions with other Chipster users
 - You can give either read-only or read-write access
- If your analysis job takes a long time, you don't need to keep Chipster open:
 - Wait that the data transfer to the server has completed (job status = running)
 - Close Chipster
 - Open Chipster later and the results will be there

Options for importing data to Chipster



- Add file button
 - Upload files
 - Upload folder
 - Download from URL
- Sessions tab
 - Import session file
- Tools
 - Import from Illumina BaseSpace
 - Utilities / Retrieve data from Illumina BaseSpace
 - Access token needed!
 - Import from SRA database
 - Utilities / Retrieve FASTQ or BAM files from SRA
 - Import from Ensembl database
 - Utilities / Retrieve data for a given organism in Ensembl
 - Import from URL
 - Utilities / Download file from URL directly to server

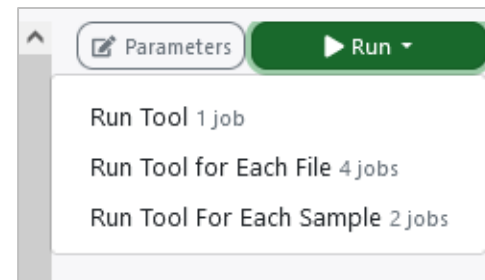
Running many analysis jobs at the same time



- You can have many analysis jobs running at the same time
 - No need to wait that one finishes before starting a new one

Run button gives several options:

- Run tool
 - Runs the selected analysis tool once
- Run tool for each file
 - Runs the selected analysis tool for each of the input files individually
- Run tool for each sample
 - If you have grouped paired end FASTQ files to samples using the Define samples –option, you can run the selected analysis tool for the input files in a sample specific manner.



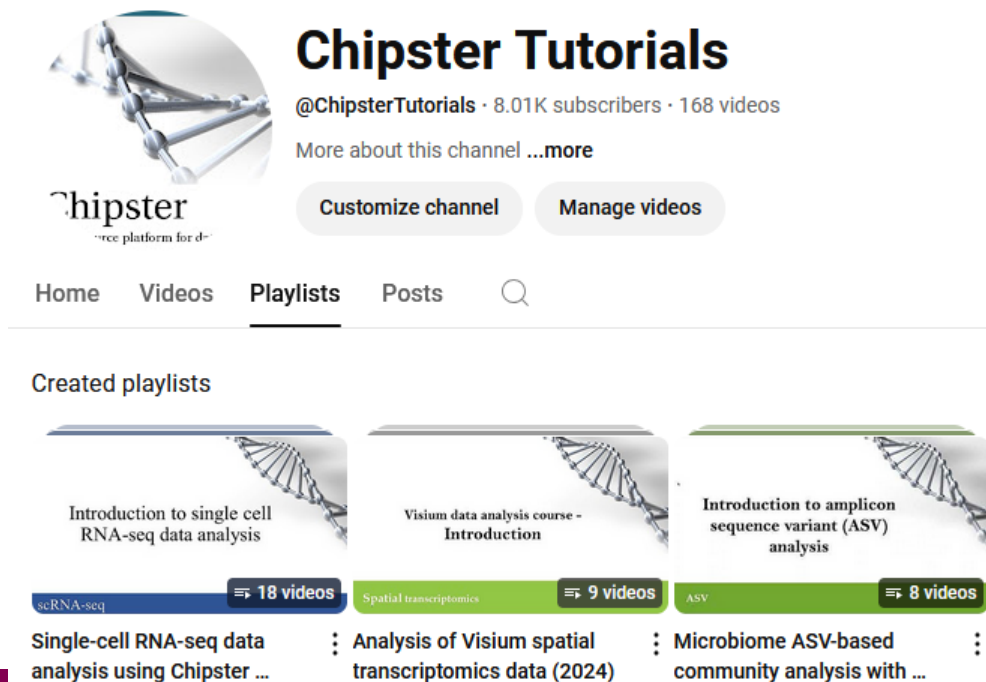
Problems? Send us a support request

-request includes the error message and (optionally) a link to your session

A screenshot of the 'Contact support' form in the Chipster application. The form is a white modal window centered on a dark grey background. The background shows a sidebar with navigation links like 'Chipster', 'Analyze', 'Sessions', 'Manual', and 'Contact'. The main content area of the background has a 'Contact support' section with a button and a 'Contact information' section with various email lists. The modal window has a title bar 'Contact support'. Inside, there is a 'Message' section with a large text input field and a red error message 'Please describe what happened'. Below that is an 'Attach session' section with two radio buttons: 'Attach a copy of your last session' (selected) and 'Don't attach the session'. The selected option has a session name 'NGS_RNAseq_fromReadsToDifferentiallyExpressedGenes_ENCODE_2samples'. Below that is a 'Your email address' section with a text input field containing 'Eija.Korpelainen@csc.fi' and a note: 'Support personnel will use this address to contact you. This email address was received from your login details. If it's not correct, please contact the organization that provided your login credentials to update it.' At the bottom right of the modal are 'Cancel' and 'Send' buttons.

More info

- chipster@csc.fi
- <http://chipster.csc.fi>
- <https://chipster.2.rahtiapp.fi/manual/courses.html>
- Chipster tutorials in YouTube

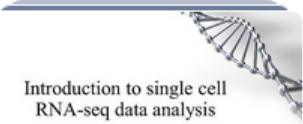
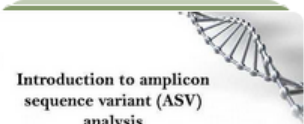


Chipster Tutorials
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Created playlists

Thumbnail	Playlist Title	Video Count	Thumbnail	Playlist Title	Video Count	Thumbnail	Playlist Title	Video Count
	Introduction to single cell RNA-seq data analysis	18 videos		Visium data analysis course - Introduction	9 videos		Introduction to amplicon sequence variant (ASV) analysis	8 videos
scRNA-seq	Single-cell RNA-seq data analysis using Chipster ...		Spatial transcriptomics	Analysis of Visium spatial transcriptomics data (2024)		ASV	Microbiome ASV-based community analysis with ...	

Introduction to microbial community / eDNA analysis

Outline

- What questions does it answer
- How is it done
- What are the main steps

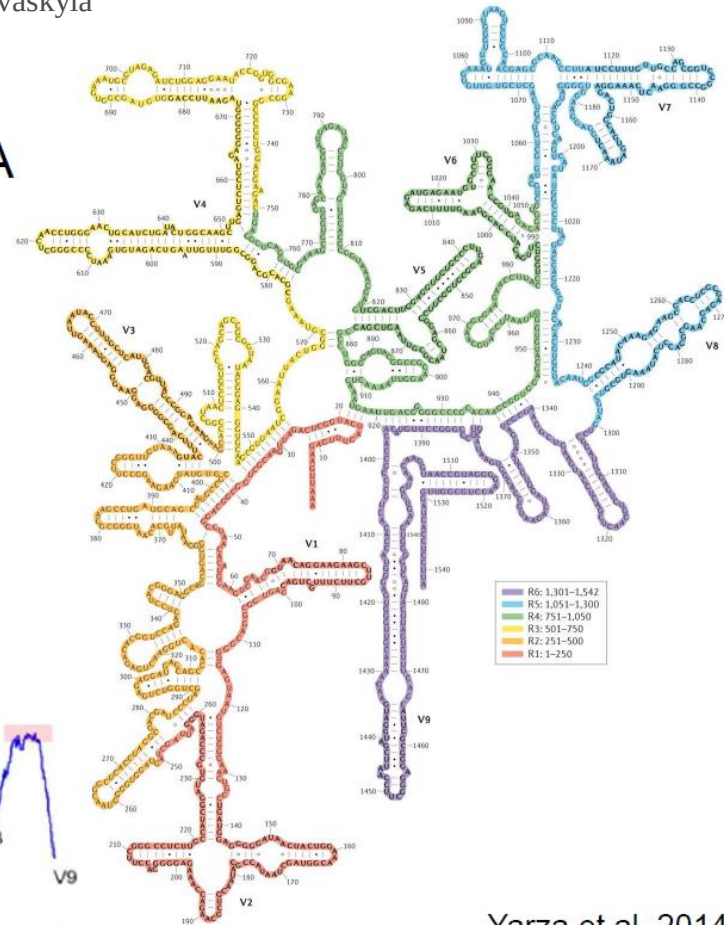
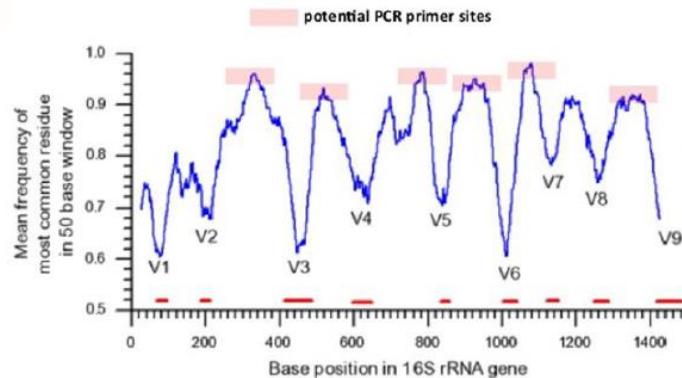
Microbial community / eDNA analysis

- Answers the questions **who are there** and **in what proportions** if compared to your other samples
 - It will not confirm that someone isn't there (sampling depth, primer/sequencing bias)
- Specific primers are used to amplify a region of one gene
 - Bacterial and archaeal communities: 16S rRNA
 - Fungal communities: ITS (internal transcribed spacer between 18S and 5.8S rRNA genes)
- Sequenced using Illumina MiSeq (or Ion Torrent)
 - New: PacBio full-length sequencing provides better resolution
- Different from metagenomics, where the aim is to sequence all genes
 - Answers the questions **who are there** and **what are they capable of doing**

Commonly used 16S rRNA gene amplicons are called by the variable regions they contain

- V1-V2
- (V3)-V4

Variable Regions of the 16S rRNA:



Yarza et al. 2014

Nature Reviews | Microbiology

UNIVERSITY OF JYVÄSKYLÄ

Main parts of microbial community analysis



- **Preprocessing**

- Quality control, trim primers/adaptors and bad quality ends
- Depending on data type:
 - MiSeq: Combine paired end reads to contigs
 - Ion Torrent: Single-end reads in one or several FASTQ files
- Filter out bad quality sequences, remove identical sequences
- Align sequences to reference template (e.g. SILVA)
- Filter sequences based on alignment position, trim sequence alignment
- Remove chimeras and sequencing errors

- **Classification and clustering**

- Taxonomic assignment of sequences (e.g. SILVA for 16S, UNITE for ITS)

- **Community analysis and visualization**

- Does community structure differ between sample groups?
- Which taxa are differentially abundant between sample groups?

How to choose the preprocessing protocol?

- Sequencing technology:
 - Illumina Miseq: paired-end short reads
 - Ion Torrent: single-end short reads
 - PacBio and Nanopore: long reads
- Gene: 16S rRNA, ITS, other?
 - reference database
 - gene characteristics
- Operational taxonomic units (OTUs) vs. amplicon sequence variants (ASVs)
 - OTUs: mothur, QIIME2
 - ASVs: DADA2

How to choose the preprocessing protocol?

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OTUs vs. ASVs

- DADA2 pipeline and ASV-based analyses hugely popular
- ASV = amplicon sequence variant
 - each sequence is a different ASV
 - different copies of 16S rRNA gene from the same organism are different ASVs
- OTU = operational taxonomic unit
 - sequences clustered into 'species' based on a similarity cut off
 - bacterial 16S rRNA gene → 97 %
 - very closely related species based on short reads cluster into the same OTU

→ clump or split?

Data set for exercises: mouse fecal samples

- Subset of 19 mouse fecal samples
- Two time points:
 - early: collected on days 0-9 post weaning
 - late: collected on days 141-150 post weaning

→ Does mouse gut microbiome change?

Data set for exercises: mouse fecal samples



- PCR amplification of V₄ region of the bacterial 16S rRNA gene (ca. 250 bp)
- Sequencing by Illumina MiSeq
- Demultiplexed (separated by sample) and barcodes removed
- Primers and sequencing adapters removed
- Each sample in two separate FASTQ files:
 - forward reads (R1)
 - reverse reads (R2)

Quality control of raw reads

Outline

- Different types of quality problems
- FASTQ file format
- Tools for checking read quality
- Tools for improving read quality

What and why?

- Potential problems
 - low confidence bases, N s
 - adapters
 - ...
- Knowing about potential problems in your data allows you to
 - correct for them before you spend a lot of time on analysis
 - take them into account when interpreting results

FASTQ file format

- Four lines per read:

@read name

GATTGTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTGTTCAACTCACAGTTT

+ read name

!"*(((***+))%%%++)(%%%%).1***-+*)"**55CCF>>>>>CCCCCCC65

- http://en.wikipedia.org/wiki/FASTQ_format
- Do **not** unzip FASTQ files, Chipster can cope with .gz files

Base qualities

- If the quality of a base is 20, the probability that it is wrong is 0.01.
 - Phred quality score $Q = -10 * \log_{10}(\text{probability that the base is wrong})$

T C A G T A C T C G

40 40 40 40 40 40 40 40 37 35

- Sanger encoding: numbers are shown as ASCII characters so that 33 is added to the Phred score
 - E.g. 39 is encoded as H, the 72nd ASCII character ($39+33 = 72$)
 - Note that older Illumina data uses different encoding

Base quality encoding systems

```

SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....

..LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....
!"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMN
|               |   |           |               |
33             59    64         73             104
0.....26...31.....40

0.2.....26...31.....41

S - Sanger          Phred+33, raw reads typically (0, 40)

L - Illumina 1.8+ Phred+33, raw reads typically (0, 41)
```

Tools for checking sequence quality

- **Read quality with MultiQC for many FASTQ files**
 - runs FastQC for all the FASTQ files simultaneously
 - checks base quality and composition, duplication, Ns, k-mers, adapters,...
 - takes a tar package of all the FASTQ files as an input file
- **Statistics for primers and adapters with TagCleaner**
 - Given an adapter or primer sequence, checks how many reads have it (allowing

tag.statistics.tsv ***

Spreadsheet Text Details

Showing all 9 rows.

#Param	Number_of_Mismatches_or_Splits	Number_of_Sequences	Percentage	Percentage_Sum
tag5	0	54996	95.61	95.61
tag5	1	2114	3.68	99.29
tag5	2	260	0.45	99.74
tag5	3	81	0.14	99.88
tag5	4	36	0.06	99.94
tag5	5	21	0.04	99.98
tag5	6	7	0.01	99.99

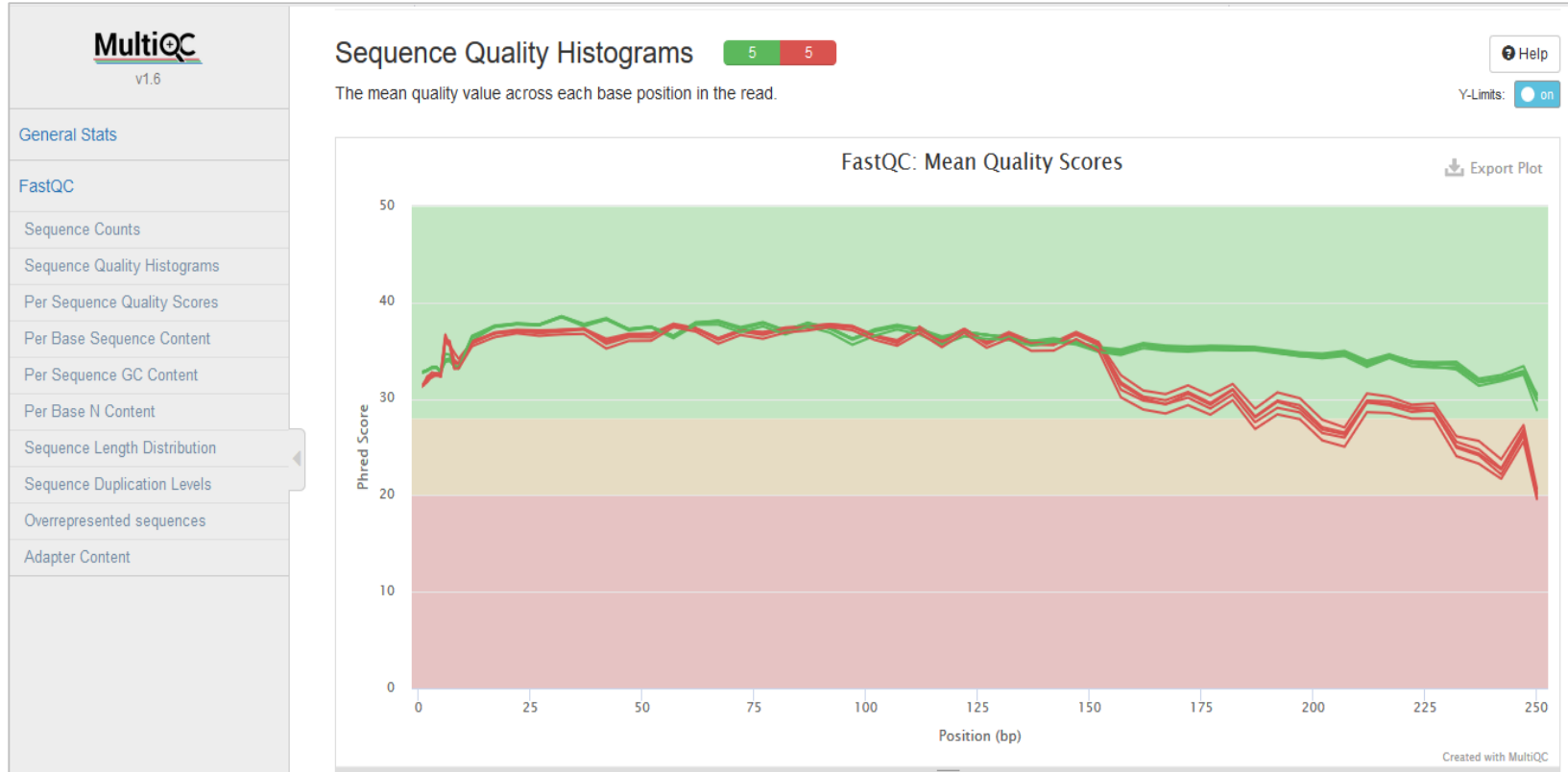
Making a Tar package of FASTQ files

- Use the tool **Utilities / Make Tar package**
- When your Tar package is ready, you can delete the original FASTQ files
 - If you want to look at the individual FASTQ files later, you can always open the Tar package using the tool **Utilities / Extract .tar.gz file**

MultiQC features

- Interactive plots
- Plots allow you to view the number or percentage of reads
- Traffic lights (they might not be suitable for your data!)
- Toolbox (click on the right side panel) allows you to
 - Highlight samples
 - Show only selected samples
 - Download plots
 - Rename samples
- Good tutorial video https://www.youtube.com/watch?v=qPbIIIO_KWNo

Per position base quality (MultiQC)



Sequence counts (MultiQC)

MultiQC
v1.7

General Stats

FastQC

Sequence Counts

Sequence Quality Histograms

Per Sequence Quality Scores

Per Base Sequence Content

Per Sequence GC Content

Per Base N Content

Sequence Length Distribution

Sequence Duplication Levels

Overrepresented sequences

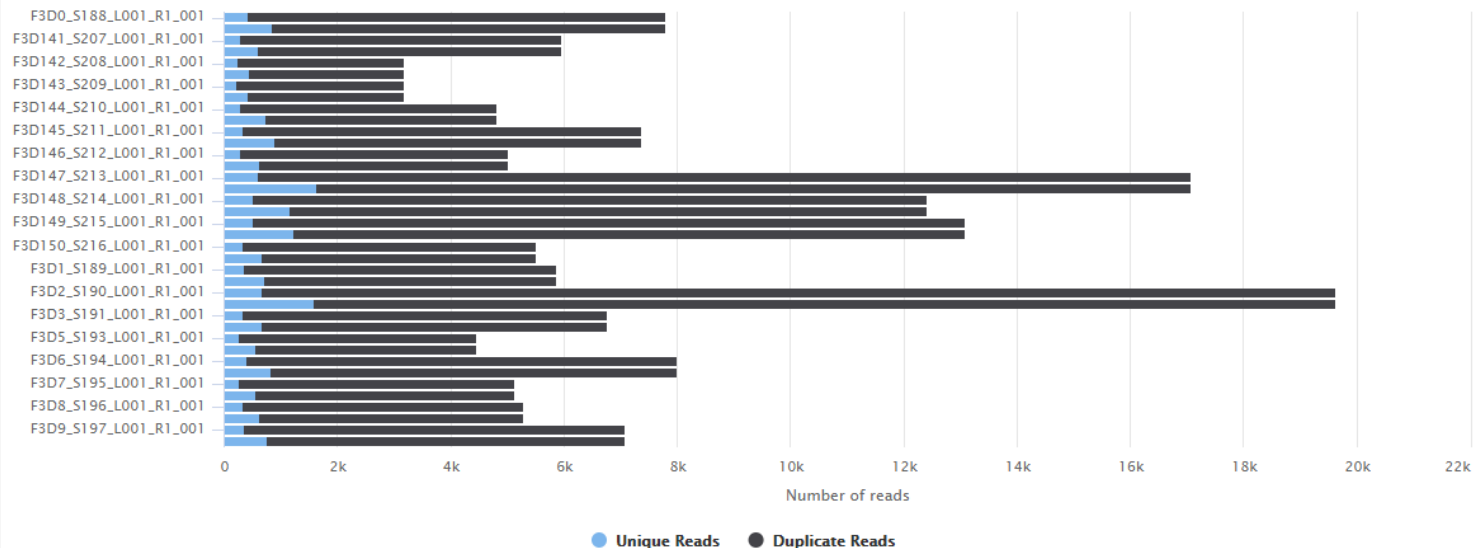
Adapter Content

Number of reads

Percentages

FastQC: Sequence Counts

Export Plot



Created with MultiQC

What if there is a quality problem?

- You can either trim or filter reads
- Filtering removes the entire read, trimming removes only the bad quality bases
 - Note that trimming can remove the entire read, if all bases are bad
- Trimming makes reads shorter, which is not always optimal
- Paired end data: the matching order of the reads in the two files has to be preserved
 - If a read is removed, its pair has to be removed as well

Preprocessing tools for improving reads

- Trimmomatic and PRINSEQ
 - Filtering based on read quality and length
 - Trimmomatic is faster
- Cutadapt
 - Removes primers and adapters allowing mismatches
- TagCleaner
 - Removes primers and adapters allowing mismatches
- FastX
 - Can be used for trimming a given number of bases from either end of the reads
 - Does not take the pairing of reads into account

Removing primers with Cutadapt

Remove primers and adapters with Cutadapt



Parameters

 Reset All

Are the reads paired end or single end

paired 

If your reads are paired end, the reverse complement of the 3' and 5' adapters will be removed from the reverse reads.

The 5' adapter:

Give here the 5 end adapter/primer.

GTCAGCTCGTGGYGTGAG



The 3' adapter:

Give here the 3 end adapter/primer in the orientation it exists in the forward/single reads, usually in reverse complement.

TTGYACACACCGCCCGT



← reverse complement

Remove reads which were not trimmed

yes 

Remove reads which did not contain an adapter.



Input files

Tar package containing the FASTQ files

zippedFastq.tar 

Combine paired reads into contigs with VSEARCH

Outline

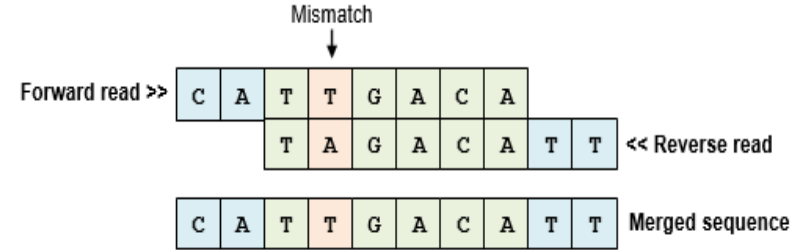
- How are paired reads merged to contigs with VSEARCH
- Things to take into account
- Posterior quality score calculations
- How to set the parameters in Chipster
- What are the output files

Combine paired reads into contigs with VSEARCH

- VSEARCH is an open source and free of charge alternative to USEARCH

Rognes et al. 2016. VSEARCH: a versatile open source tool for metagenomics. *PeerJ* 4:e2584

- Input file:** Tar package of FASTQ files
- Creates a reverse complement of the reverse read
- Extracts the sequences and the quality data from the FASTQ files and joins the paired reads to contigs.
- Calculates also the posterior Quality scores.



Usearch fastq_mergepairs manual:
https://www.drive5.com/usearch/manual/merge_pair.html

Posterior quality scores

- If the base calls match, the Q score should increase
- If there is a mismatch, Q score should usually reduce
- Uses Baye's Theorem to calculate posterior Q scores:
 - R.C. Edgar and H. Flyvbjerg (2015), "Error filtering, pair assembly and error correction for next-generation sequencing reads", Bioinformatics 31(21) 3476-3482,

Forward read										
C	A	T	T	G	A	C	A			
32	34	20	20	28	16	14	10			
Reverse read		T	A	G	A	C	A	T	T	Base calls
		2	5	4	8	12	20	38	40	Q scores
		C	A	T	T	G	A	C	A	Consensus
		32	34	22	16	35	28	30	34	Posterior Qs

Parameters

Combine paired reads to contigs with VSEARCH

Parameters

Reset All

Maximum number of non-matching nucleotides

10

The maximum number of non-matching nucleotides allowed in the overlap region 0 - 100, the default value is 10

Maximum percentage of non-matching nucleotides

100

The maximum percentage of non-matching nucleotides allowed in the overlap region. The default value is 100.0%

Discard input sequences with more than this number of Ns.

Discard input sequences which contain with more than the specified number of ambiguous bases

Input files

Tar package containing the FASTQ files

chipster.tar

List of FASTQ files by sample

Run options



Run Tool

Runs the tool once. Uses the one selected file as the tool input.



Run Tool for Each File

Only one file selected.



Run Tool for Each Sample

Only one file selected.

Close

Output files:

- contigs.tar
 - tar package containing all the merged FASTQ files
- summary.txt
- summary_stats.tsv
- samples.fastqs.txt

Samples.fastqs.txt

- Check if the FASTQ files were assigned correctly to each sample
- If the assignment is wrong, make this file yourself and give it as input

F3D0	F3D0_S188_L001_R1_001.fastq.gz	F3D0_S188_L001_R2_001.fastq.gz
F3D141	F3D141_S207_L001_R1_001.fastq.gz	F3D141_S207_L001_R2_001.fastq.gz
F3D142	F3D142_S208_L001_R1_001.fastq.gz	F3D142_S208_L001_R2_001.fastq.gz
F3D143	F3D143_S209_L001_R1_001.fastq.gz	F3D143_S209_L001_R2_001.fastq.gz
F3D144	F3D144_S210_L001_R1_001.fastq.gz	F3D144_S210_L001_R2_001.fastq.gz
F3D145	F3D145_S211_L001_R1_001.fastq.gz	F3D145_S211_L001_R2_001.fastq.gz
F3D146	F3D146_S212_L001_R1_001.fastq.gz	F3D146_S212_L001_R2_001.fastq.gz
F3D147	F3D147_S213_L001_R1_001.fastq.gz	F3D147_S213_L001_R2_001.fastq.gz
F3D148	F3D148_S214_L001_R1_001.fastq.gz	F3D148_S214_L001_R2_001.fastq.gz
F3D149	F3D149_S215_L001_R1_001.fastq.gz	F3D149_S215_L001_R2_001.fastq.gz
F3D150	F3D150_S216_L001_R1_001.fastq.gz	F3D150_S216_L001_R2_001.fastq.gz
F3D1	F3D1_S189_L001_R1_001.fastq.gz	F3D1_S189_L001_R2_001.fastq.gz
F3D2	F3D2_S190_L001_R1_001.fastq.gz	F3D2_S190_L001_R2_001.fastq.gz
F3D3	F3D3_S191_L001_R1_001.fastq.gz	F3D3_S191_L001_R2_001.fastq.gz
F3D5	F3D5_S193_L001_R1_001.fastq.gz	F3D5_S193_L001_R2_001.fastq.gz
F3D6	F3D6_S194_L001_R1_001.fastq.gz	F3D6_S194_L001_R2_001.fastq.gz
F3D7	F3D7_S195_L001_R1_001.fastq.gz	F3D7_S195_L001_R2_001.fastq.gz
F3D8	F3D8_S196_L001_R1_001.fastq.gz	F3D8_S196_L001_R2_001.fastq.gz
F3D9	F3D9_S197_L001_R1_001.fastq.gz	F3D9_S197_L001_R2_001.fastq.gz

Summary_stats.tsv

Summary table based on the data in summary.txt to help to find information about merging

Showing all 19 rows.

Sample	Pairs	Merged	Not_merged	Percentage_not_merged	Mean_expected_error
F3D0	7793	6596	1197	15.4	0.08
F3D141	5958	4903	1055	17.7	0.08
F3D142	3183	2546	637	20.0	0.09
F3D143	3178	2610	568	17.9	0.09
F3D144	4827	3695	1132	23.5	0.10
F3D145	7377	5666	1711	23.2	0.09
F3D146	5021	4177	844	16.8	0.08
F3D147	17070	12816	4254	24.9	0.09
F3D148	12405	9952	2453	19.8	0.08
F3D149	13083	10774	2309	17.6	0.08
F3D150	5509	4410	1099	19.9	0.09
F3D1	5869	5066	803	13.7	0.09
F3D2	19620	16566	3054	15.6	0.08
F3D3	6758	5490	1268	18.8	0.08
F3D5	4448	3635	813	18.3	0.09
F3D6	7989	6788	1201	15.0	0.08
F3D7	5129	4263	866	16.9	0.08
F3D8	5294	4599	695	13.1	0.08
F3D9	7070	6148	922	13.0	0.08

Summary.txt

Merge information on every sample

```
# F3D0
```

```
vsearch v2.17.1_linux_x86_64, 338.3GB RAM, 80 cores  
https://github.com/torognes/vsearch
```

```
Merging reads 100%
```

```
7793 Pairs  
6596 Merged (84.6%)  
1197 Not merged (15.4%)
```

```
Pairs that failed merging due to various reasons:
```

- 1 too few kmers found on same diagonal
- 2 multiple potential alignments
- 602 too many differences
- 2 too high percentage of differences
- 588 alignment score too low, or score drop too high
- 2 staggered read pairs

```
Statistics of all reads:
```

```
250.83 Mean read length
```

```
Statistics of merged reads:
```

```
252.52 Mean fragment length  
0.53 Standard deviation of fragment length  
0.43 Mean expected error in forward sequences  
1.05 Mean expected error in reverse sequences  
0.08 Mean expected error in merged sequences  
0.40 Mean observed errors in merged region of forward sequences  
1.67 Mean observed errors in merged region of reverse sequences  
2.07 Mean observed errors in merged region
```

Filter sequences based on the number of expected errors



Outline:

- How to filter contigs based on the number of expected errors
- How to calculate expected errors
- What are the output files

Filter sequences based on the number of expected errors



- Based on the `fastq_filter` command in VSEARCH package
- Input file: Tar package of FASTQ files
- Use posterior quality scores to calculate the average number of errors.
 - Better prediction of the true quality scores after merging.
- Small E means high quality, high E means low quality

summary.tsv:

Parameters:

- Discard sequences with more than the specified number of expected errors
 - decimal number, default 1

Output files:

- filtered_contigs.tar
- summary.tsv

Showing all 20 rows.

Sample	Sequences_kept	Sequences_discarded	Percentage_of_discarded
All	119194	1507	
F3D0	6512	84	1.2735
F3D141	4845	58	1.1829
F3D142	2510	36	1.414
F3D143	2580	30	1.1494
F3D144	3628	67	1.8133
F3D145	5594	72	1.2707
F3D146	4119	58	1.3886
F3D147	12657	159	1.2406
F3D148	9867	85	0.8541
F3D149	10640	135	1.2529
F3D150	4352	58	1.3152
F3D1	4999	67	1.3225
F3D2	16346	220	1.328
F3D3	5435	55	1.0018
F3D5	3569	66	1.8157
F3D6	6701	87	1.2817
F3D7	4221	42	0.9852
F3D8	4540	59	1.2829
F3D9	6079	69	1.1223

Combine files and make a count file

- Use the tool **Combine FASTQ files into one FASTA file and make a Mothur count file** to

- converts FASTQ to FASTA
- merges all the samples in one file
- creates the Mothur count file

count file = keeping track which sequence belongs to which sample

sequences.count_table ...

Spreadsheet [Text](#) [Open in New Tab](#) [Details](#)

Showing the first 100 rows. View in [full screen](#) to see all rows and total row count.

Representative_Sequence	total	HPc1_cut	HPc2_cut	HPc5_cut	HPc6_cut	HPps1_cut	HPps2_cut
3UBKS_00109_00128	1	0	0	0	0	0	0
3UBKS_00116_02153	1	0	0	0	0	0	0
3UBKS_00121_00096	1	0	0	1	0	0	0
3UBKS_00134_02237	1	0	0	1	0	0	0
3UBKS_00142_02256	1	0	0	0	0	0	0
3UBKS_00146_00115	1	0	1	0	0	0	0
3UBKS_00148_02235	1	0	1	0	0	0	0
3UBKS_00149_00032	1	0	0	0	0	0	0
3UBKS_00152_00208	1	0	1	0	0	0	0
3UBKS_00167_00138	1	0	1	0	0	0	0
3UBKS_00168_00271	1	0	0	0	0	0	0
3UBKS_00171_00252	1	0	0	0	0	0	0
3UBKS_00172_02273	1	1	0	0	0	0	0

Screen reads for ambiguous bases and too long reads

- Tool **Screen sequences for several criteria**
- **Input:** FASTA file and count file (ends .count_table)

Same tool used later for screening sequence alignment!

Screen sequences for several criteria

Parameters
Reset All

Maximum number of ambiguous bases
How many ambiguous bases are allowed in a sequence
0

Maximum homopolymer length
Maximum length of homopolymers allowed

Minimum length
What is the minimum length of the sequences to be kept?

Maximum length
What is the maximum length of the sequences to be kept?
400

Alignment start position
Remove sequences which start after this position

Alignment end position
Remove sequences which end before this position

Optimize by
Optimize according to minlength, start or end position. Please note that if you use this option, you can't determine the same criteria above! Fill in the optimization criteria below as well.
empty

Optimization criteria
Optimization criteria. For example 85 means that Mothur will optimize the cutoff for the above chosen quality so that 85% of the sequences are kept.

Input files

FASTA file
sequences.fasta.gz

Groups file
No compatible files

Count file
sequences.count_table

Remove identical sequences

- The FASTA file contains many identical sequences
- Aligning the same sequence to the reference would be computationally wasteful
- We remove identical sequences and keep only one representative in the FASTA file
 - count file keeps track of how many sequences the representative represents in the samples
- Tool **Extract unique sequences**
- Give fasta file and count file as **input**
- Output files
 - unique.fasta = unique sequences
 - unique.count_table = how many represented sequences are in each sample
 - unique.summary.tsv = sequence information

unique.summary.tsv

- Number of sequences: total and unique
- Stats (min, max, mean, median and quantiles) of
 - number of bases
 - number of ambiguous bases
 - start and end positions
 - homopolymer length

unique.summary.tsv ...

Spreadsheet

Text

Open in New Tab

Details

Showing all 10 rows.

<empty>	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	200	200	0	3	1
2.5%-tile:	1	201	201	0	4	2782
25%-tile:	1	215	215	0	4	27817
Median:	1	224	224	0	4	55634
75%-tile:	1	257	257	0	5	83450
97.5%-tile:	1	307	307	0	6	108485
Maximum:	1	333	333	0	8	111266
Mean:	1	236	236	0	4	
# of unique seqs:	54053					
total # of seqs:	111266					

Output so far:

- FASTA file: unique.fasta.gz
 - trimmed and filtered unique sequence reads
 - .gz indicates file compression
- Count file: unique.count_table
 - which sequence belongs to which sample
- Next: alignment

unique.count_table ...

Spreadsheet Text Open in New Tab Details

Showing the first 100 rows. View in [full screen](#) to see all rows and total row count.

Representative_Sequence	total	F3D0	F3D1	F3D141	F3D142	F
M00967_43_000000000-A3JHG_1_1101_14069_1827	4443	373	27	263	147	1
M00967_43_000000000-A3JHG_1_1101_18044_1900	27	1	0	1	0	0
M00967_43_000000000-A3JHG_1_1101_13234_1983	9684	402	268	312	193	1
M00967_43_000000000-A3JHG_1_1107_8348_3774	675	24	39	31	3	9
M00967_43_000000000-A3JHG_1_1101_16780_2259	174	14	17	1	7	3
M00967_43_000000000-A3JHG_1_1107_21258_3786	11	0	0	2	0	1
M00967_43_000000000-A3JHG_1_1101_18089_2781	591	43	108	4	0	1
M00967_43_000000000-A3JHG_1_1107_14811_3878	1	0	0	0	0	0
M00967_43_000000000-A3JHG_1_1101_14203_2907	5889	309	131	219	102	1
M00967_43_000000000-A3JHG_1_1107_17052_3882	3	0	1	0	0	0
M00967_43_000000000-A3JHG_1_1107_14125_3942	117	3	2	10	3	6
M00967_43_000000000-A3JHG_1_1101_18278_3345	3	2	0	0	0	1

Align sequences to reference template alignment

Outline

- SILVA reference template alignment
- Alignment steps
- How to improve and speed up the alignment
- Alignment file format

SILVA reference template alignment

- To identify the sequences we align them to a reference template alignment
- Chipster uses the full SILVA template, but you can also give your own
- The current SILVA version in Chipster is 138.1 (version 138.2 coming soon)
 - Contains 146 601 sequences: 128 884 bacteria, 2846 archaea, and 14 871 eukarya
 - the alignment is 50 000 columns long so that it is compatible with 18S rRNA sequences and archaeal 16S rRNA sequences
 - to make alignment process faster, indicate which region of the SILVA template alignment matches the area you amplified
 - to get the SILVA coordinates of that area, align a small number of samples first
- https://mothur.org/wiki/Silva_reference_files

Aligning sequences to template alignment

- Tool **Align sequences to reference**, based on Mothur align.seqs and pcr.seqs commands
- Give unique.fasta.gz and unique.count_table as **input**
- Three steps
 - find the closest template sequence for the query sequence using K-mer search with 8mers
 - align the query and the de-gapped template sequence using Needleman-Wunsch pairwise alignment
 - re-insert gaps to the query and template pairwise alignment using the NAST algorithm so that the query sequence alignment is compatible with the original template alignment
- Speed depends on the number and length of the query and template sequences
- Limit the alignment to the template region which corresponds to the part of the 16S rRNA gene you amplified → better alignment quality, less space needed

Result files

- aligned.fasta.gz = aligned sequences
 - periods lead to the first base in the sequence and follow the last base of the sequence
- custom.reference.summary.tsv = information on the region of the reference used
- aligned-summary.tsv = aligned sequence information

aligned.fasta ***

[Text](#) [Details](#)

First 100.0 kB. View in [full screen](#) to see the whole 146.2 MB file.

```
>M00967_43_000000000-A3JHG_1_1107_25112_15468
.....AC---GG-AG-GAT-----
>M00967_43_000000000-A3JHG_1_2104_24218_17682
.....AC---GG-AG-GAT-----
>M00967_43_000000000-A3JHG_1_2111_10309_12747
.....AC---GG-AG-GAT-----
>M00967_43_000000000-A3JHG_1_1113_16474_12480
.....AC---GT-AG-GGG-----
>M00967_43_000000000-A3JHG_1_2113_18674_18253
.....AC---GG-AG-GAT-----
>M00967_43_000000000-A3JHG_1_1111_26127_23565
.....AC---GT-AG-GGG-----
>M00967_43_000000000-A3JHG_1_2102_6927_12866
.....AC---GT-AG-GTG-----
>M00967_43_000000000-A3JHG_1_2110_12809_22465
.....AC---GG-AG-GAT-----
>M00967_43_000000000-A3JHG_1_1108_20705_14486
```

aligned-summary.tsv ***

[Spreadsheet](#) [Text](#) [Details](#)

Showing all 10 rows.

<empty>	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	8	8715	248	0	3	1
2.5%-tile:	8	9582	251	0	3	3119
25%-tile:	8	9582	251	0	4	31185
Median:	8	9582	251	0	4	62370
75%-tile:	8	9582	252	0	5	93555
97.5%-tile:	8	9582	252	0	6	121621
Maximum:	1801	9582	255	0	11	124739
Mean:	8	9581	251	0	4	
# of unique seqs:	15920					
total # of seqs:	124739					

custom.reference.summary.tsv ***

[Spreadsheet](#) [Text](#) [Details](#)

Showing all 10 rows.

<empty>	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	7908	44	0	3	1
2.5%-tile:	8	9582	250	0	3	3666
25%-tile:	8	9582	252	0	4	36651
Median:	8	9582	252	0	5	73301
75%-tile:	8	9582	252	0	5	109951
97.5%-tile:	8	9582	419	0	6	142936
Maximum:	2425	9582	1081	5	16	146601
Mean:	8	9581	269	0	4	
# of Seqs:	146601					

Filter and trim aligned sequences

Outline

- Filter sequences based on alignment start and end position
- Trim sequence alignment
- Remove identical sequences

Filter aligned sequences

- All the aligned sequences should overlap the same alignment coordinates
- Remove deviants by filtering based on the alignment start and end position
 - Check aligned-summary.tsv
- Remove also sequences which have homopolymers longer than those in the reference
 - Check custom.reference.summary.tsv
- Tool: **Screen sequences for several criteria** (based on Mothur command screen.seqs)
- **Input files:** aligned.fasta.gz and unique.count_table
- **Result files**
 - screened.fasta.gz = screened sequences
 - screened.count_table = updated count_table
 - summary.screened.tsv = sequence information

Parameters for filtering aligned sequences

Screen sequences for several criteria



Parameters

Maximum number of ambiguous bases

How many ambiguous bases are allowed in a sequence

Maximum homopolymer length

Maximum length of homopolymers allowed



Minimum length

What is the minimum length of the sequences to be kept?

Maximum length

What is the maximum length of the sequences to be kept?

Alignment start position

Remove sequences which start after this position



Alignment end position

Remove sequences which end before this position



Optimize by

Optimize according to minlength, start or end position. Please note that if you use this option, you can't determine the same criteria above! Fill in the optimization criteria below as well.

Optimization criteria

Optimization criteria. For example 85 means that Mothur will optimize the cutoff for the above chosen quality so that 85% of the sequences are kept.

Input files

FASTA file

Groups file

Count file

Trim sequence alignment for overhangs and empty columns



- We remove overhangs (columns containing .) and keep the common alignment region
- Gap columns (where all the characters are –) have no information, so we remove them
 - makes distance calculation faster
- Removing alignment columns can create identical sequences → need to remove them
- Tool **Filter sequence alignment** (based on Mothur commands filter.seqs and unique.seqs)

• and screened



```
.....T-----AC---GG-AG-GAT-----
.....T-----AC---GG-AG-GAT-----
.....T-----AC---GG-AG-GAT-----
.....T-----AC---GT-AG-GTG-----
.....T-----AC---GG-AG-GAT-----
.....T-----AC---GT-AG-GGG-----
```

```
>M00967_43_000000000-A3JHG_1_1101_14069_1827
TAC--GG-AG-GAT--GCG-A-G-C-G-T-T--AT-C-CGG-AT--TT-A-T-T-
>M00967_43_000000000-A3JHG_1_1101_18044_1900
TAC--GG-AG-GAT--GCG-A-G-C-G-T-T--GT-C-CGG-AA--TC-A-C-T-
>M00967_43_000000000-A3JHG_1_1101_13234_1983
TAC--GG-AG-GAT--GCG-A-G-C-G-T-T--AT-C-CGG-AT--TT-A-T-T-
>M00967_43_000000000-A3JHG_1_1101_16780_2259
TAC--GT-AG-GTG--GCA-A-G-C-G-T-T--AT-C-CGG-AT--TT-A-C-T-
>M00967_43_000000000-A3JHG_1_1101_17674_2779
TAC--GG-AG-GAT--GCG-A-G-C-G-T-T--AT-C-CGG-AT--TT-A-T-T-
>M00967_43_000000000-A3JHG_1_1101_18089_2781
TAC--GT-AG-GGG--GCA-A-G-C-G-T-T--AT-C-CGG-AT--TT-A-C-T-
```

Result files

- filtered-unique.fasta.gz = trimmed aligned sequences
- filtered-unique.count_table = updated count_table
- filtered-unique-summary.tsv = sequence information
- filtered-log.txt = how many alignment columns were removed

filtered-log.txt ...

Text [Details](#)

File size 158.0 bytes.

```
Length of filtered alignment: 366
Number of columns removed: 9216
Length of the original alignment: 9582
Number of sequences used to construct filter: 15800
```

Remove sequencing errors and chimeras

Outline

- How preclustering works
- What are chimeras and how to remove them?

Precluster very similar sequences

- Assumes that abundant sequences are more likely to generate sequencing errors
 - ranks sequences in order of their abundance
 - walks through the list looking for rarer sequences which differ only by x number of bases from the original sequence (allow 1 mismatch for every 100 bp of sequence)
 - merges those that are within the threshold
- Tool: **Precluster aligned sequences** (based on Mothur command precluster.seqs)
- **Input files:** filtered-unique.fasta.gz and filtered-unique.count_table
- **Result files**
 - preclustered.fasta.gz = preclustered aligned sequences
 - preclustered.count_table = updated count_table
 - preclustered-summary.tsv = sequence information

Remove chimeras



- Chimera = artifact sequence formed by two biological sequences
 - incomplete extension during PCR allows subsequent PCR cycles to use a partially extended strand to bind to the template of a similar sequence.
 - the partially extended strand then acts as a primer to extend and form a chimeric sequence.
 - as many as 30% of the sequences from mixed template environmental samples may be chimeric.
- Tool: **Remove chimeric sequences** (based on Mothur chimera.uchime, chimera.vsearch)
- You can either use a reference or detect chimeras *de novo*
 - Reference is the bacterial subset of the Silva Gold 16S rRNA
 - *De novo* approach uses the more abundant sequences in your data as the reference
- Dereplicate = should we remove a chimera only from the sample where it was spotted?
 - True = only from that sample (“do not replicate”)
 - False = from all samples (“replicate to other samples”)

• **Input files:** preclustered.fasta.gz and preclustered.count_table file

Chimera removal results

- Result files
 - `chimeras.removed.fasta.gz` = aligned sequences
 - `chimeras.removed.count_table` = updated `count_table`
 - `chimeras.removed.summary.tsv` = sequence information
- Results and running time depend on the method and reference used. Example:
 - 6022 unique sequences to start with
 - 5283 after chimera removal with VSEARCH and SILVA gold (29 s)
 - 2467 after chimera removal with VSEARCH and *de novo* (4 s)
 - 5323 after chimera removal with UCHIME and SILVA gold (23 min)
 - 5023 after chimera removal with UCHIME and *de novo* (19 s)

Classify sequences to taxonomic units

Outline

- Tools for assigning sequences to taxonomies
- Wang method
- File formats
 - Taxonomy assignment file
 - Classification summary file

Assign sequences to taxonomy

- Tools **Classify 16S or 18S sequences to taxonomic units using Silva** and **Classify ITS sequences to taxonomic units using UNITE** (based on Mothur command classify.seqs)
 - you can also use your own taxonomy by providing reference fasta and taxonomy outline file.
- Wang method
 - looks at the query sequence k-mer by k-mer
 - calculates the probability a sequence from a given taxonomy would contain a specific k-mer
 - calculates the probability a query sequence would be in a given taxonomy based on the k-mers it contains, and assigns the query sequence to the taxonomy with the highest probability
 - calculates bootstrap confidence score for the assignment (chooses randomly $1/8$ of the k-mers in the query), by default 100 iterations
 - if the confidence is <80%, assignment will revert to higher level
- **Input files:** chimeras.removed.fasta.gz and chimeras.removed.count_table

Classification result files



- sequences-taxonomy-assignment.txt = sequence name and taxonomy

sequences-taxonomy-assignment.txt ...

Text Details

First 100.0 kB. View in [full screen](#) to see the whole 826.1 kB file. [Full Screen](#)

```

M00967_43_000000000-A3JHG_1_2108_3323_12495  Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(99);Muribaculaceae_ge(99);
M00967_43_000000000-A3JHG_1_1110_13281_5144  Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100);Muribaculaceae_ge(100);
M00967_43_000000000-A3JHG_1_1104_14019_23841  Bacteria(100);Firmicutes(100);Clostridia(100);Lachnospirales(100);Lachnospiraceae(100);Lachnospiraceae_NK4A136_group(94);
M00967_43_000000000-A3JHG_1_2101_16841_5683  Bacteria(100);Firmicutes(100);Clostridia(100);Lachnospirales(100);Lachnospiraceae(100);ASF356(100);
M00967_43_000000000-A3JHG_1_1113_12062_8139  Bacteria(100);Firmicutes(100);Clostridia(100);Oscillospirales(100);Ruminococcaceae(100);Incertae_Sedis(97);
M00967_43_000000000-A3JHG_1_2103_24284_21559  Bacteria(100);Firmicutes(100);Clostridia(100);Lachnospirales(100);Lachnospiraceae(100);Lachnospiraceae_unclassified(100);
M00967_43_000000000-A3JHG_1_2109_25056_22893  Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100);Muribaculaceae_ge(98);
M00967_43_000000000-A3JHG_1_2113_12029_24665  Bacteria(100);Firmicutes(90);Clostridia(89);Clostridia_vadinBB60_group(87);Clostridia_vadinBB60_group_fa(87);Clostridia_vadinBB60_group_ge(87);
M00967_43_000000000-A3JHG_1_2110_11085_20271  Bacteria(100);Firmicutes(100);Clostridia(100);Clostridiales(100);Clostridiaceae(100);Clostridium_sensu_stricto_1(86);
    
```

- classification-summary.tsv = the number of sequences that were found at each level

classification-summary.tsv ...

Spreadsheet Text Details

Showing the first 100 of 208 rows. View in [full screen](#) to see all rows. [Full Screen](#)

taxlevel	rankID	taxon	daughterlevels	total	F3D0	F3D1	F3D141	F3D142	F3D143	F3D144	F3D145	F3D146	F3D147	F3D148	F3D149	F3D150	F3D2	F3D3	F3D4
0	0	Root	1	123474	6573	4906	5046	2634	2636	3842	6063	4246	14014	10415	10962	4558	16667	5628	368
1	0.1	Bacteria	9	123474	6573	4906	5046	2634	2636	3842	6063	4246	14014	10415	10962	4558	16667	5628	368
2	0.1.1	Actinobacteriota	2	400	27	4	20	29	13	28	10	9	50	81	43	28	19	20	3
3	0.1.1.1	Actinobacteria	3	265	23	0	16	28	10	21	7	3	29	68	25	16	4	15	0
4	0.1.1.1.1	Actinomycetales	1	2	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0
5	0.1.1.1.1.1	Actinomycetaceae	1	2	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0
6	0.1.1.1.1.1.1	Actinomyces	0	2	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0
4	0.1.1.1.2	Bifidobacteriales	1	262	23	0	16	27	10	21	5	3	29	68	25	16	4	15	0
5	0.1.1.1.2.1	Bifidobacteriaceae	2	262	23	0	16	27	10	21	5	3	29	68	25	16	4	15	0

Removing unwanted lineages

- Data may contain assignments to mitochondria, chloroplasts, unknown
- You can remove these after converting Mothur files into phyloseq object
 - **Remove selected taxa** removes chloroplast and mitochondrial sequences from a phyloseq object, and up to five user-specified taxa at the desired level of biological organization.
 - **Filter by taxonomic group** tidies a phyloseq object so that OTUs only from the desired taxonomic group (bacteria, archaea, eukaryotes or fungi) are retained. Features with ambiguous phylum-level annotation (e.g. NA, unknown, uncharacterized) are removed.

selected.txt ***

Text Details

File size 1.4 kB.

Full Screen

M00967_43_000000000-A3JHG_1_2114_13761_23520	Bacteria(100);Proteobacteria(100);Alphaproteobacteria(100);Rickettsiales(100);Mitochondria(100);Mitochondria_ge(100);
M00967_43_000000000-A3JHG_1_1104_7270_11276	Bacteria(100);Proteobacteria(100);Alphaproteobacteria(100);Rickettsiales(100);Mitochondria(100);Mitochondria_ge(100);
M00967_43_000000000-A3JHG_1_2113_17555_7199	Bacteria(100);Proteobacteria(100);Alphaproteobacteria(100);Rickettsiales(100);Mitochondria(100);Mitochondria_ge(100);
M00967_43_000000000-A3JHG_1_2113_14852_17911	Bacteria(100);Proteobacteria(100);Alphaproteobacteria(100);Rickettsiales(100);Mitochondria(100);Mitochondria_ge(100);
M00967_43_000000000-A3JHG_1_2114_8404_6272	Bacteria(100);Proteobacteria(100);Alphaproteobacteria(100);Rickettsiales(100);Mitochondria(100);Mitochondria_ge(100);

selected.txt ***

Text Details

File size 1.5 kB.

Full Screen

M00967_43_000000000-A3JHG_1_2113_12328_2096	Bacteria(100);Cyanobacteria(100);Cyanobacteriia(100);Chloroplast(100);Chloroplast_fa(100);Chloroplast_ge(100);
M00967_43_000000000-A3JHG_1_1108_25652_13962	Bacteria(100);Cyanobacteria(100);Cyanobacteriia(100);Chloroplast(100);Chloroplast_fa(100);Chloroplast_ge(100);
M00967_43_000000000-A3JHG_1_2103_14400_26861	Bacteria(100);Cyanobacteria(100);Cyanobacteriia(100);Chloroplast(100);Chloroplast_fa(100);Chloroplast_ge(100);
M00967_43_000000000-A3JHG_1_1108_15071_6951	Bacteria(100);Cyanobacteria(100);Cyanobacteriia(100);Chloroplast(100);Chloroplast_fa(100);Chloroplast_ge(100);
M00967_43_000000000-A3JHG_1_1107_5566_13866	Bacteria(100);Cyanobacteria(100);Cyanobacteriia(100);Chloroplast(100);Chloroplast_fa(100);Chloroplast_ge(100);

Fungal ITS data analysis – what is different?



- **Preprocessing**

- Quality control, trim primers/adaptors
- Combine paired end reads to contigs
- Filter out bad quality sequences, remove identical sequences
- ~~Align sequences to reference template (e.g. SILVA)~~
- ~~Filter sequences based on alignment position, trim sequence alignment~~
- Remove chimeras and sequencing errors

- **Classification and clustering**

- Taxonomic assignment of sequences using the UNITE reference
- ! When running **Cluster sequences to OTUs and classify them** set **Type of data = ITS**
→ AGC instead of OptiClust is used for clustering, because the sequences are not aligned)

- **Community analysis and visualization**

Processing data of other genes than 16S, 18S or ITS

- Similar pre-processing workflows can be followed as with 16S, 18S or ITS data, but you have to provide the following files:
 - OTU workflow: Mothur-formatted reference sequence FASTA file and taxonomy file
 - ASV workflow: DADA2-formatted reference sequence FASTA file
- Depending on the gene, you might have to skip the alignment step or align the sequences outside Chipster.

ASV-based analysis and DADA2 in Chipster

- Pre-processing: Microbial amplicon data preprocessing for ASV
 - Remove primers and adapters
 - Filter and trim
 - Sample inference
 - Combine paired reads to contigs
 - Remove chimeras
 - Assign taxonomy
- ➡ Check videos on the Chipster Youtube channel for details
- ➡ All DADA2 materials:
<https://chipster.2.rahtiapp.fi/manual/courses.html#microbiome>

Outline of Day 1:

Quality check



(Remove primers)



Pair reads into contigs



Remove identical sequences



Alignment



Filter and trim alignment



Precluster

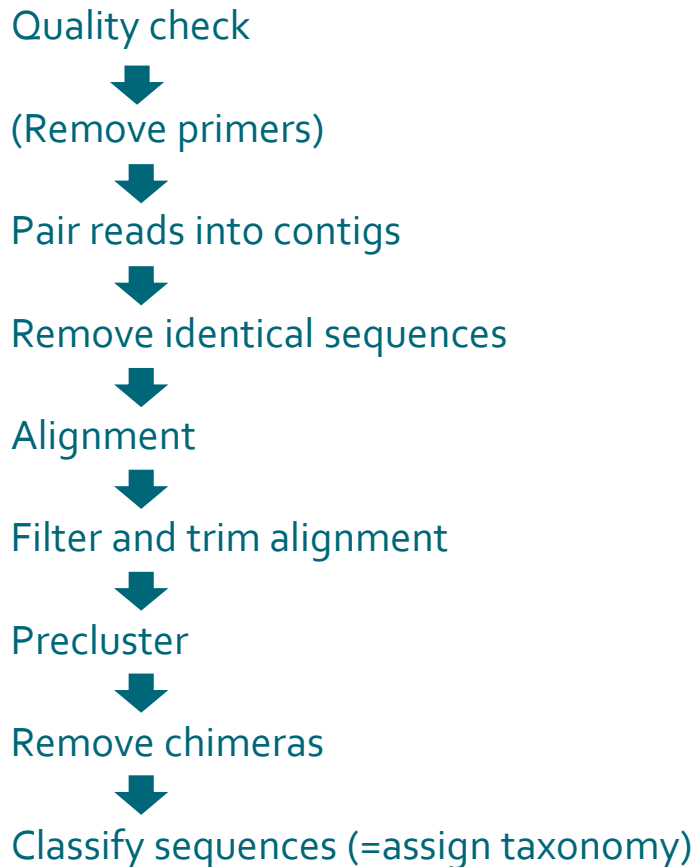


Remove chimeras



Classify sequences (=assign taxonomy)

Outline of Day 1:



Output so far:

1. FASTA file of processed reads
2. count file (which read in which sample)
3. taxonomy file (taxonomy of each read)



Day 2:

- Clustering into OTUs
- **Phyloseq** object with sample data
- Data tidying & transformations
- Taxonomy plots
- Alpha diversity
- Beta diversity: ordinations & statistics